Differentiate each of the following:

1. $y=(2x+1)^5$
2. $y=(1-x)^3$
3. $y=(x^3-2x)^2$
4. $f(x)=\dfrac{1}{1-x}$
5. $f(x)=\dfrac{1}{(2x+3)^2}$
6. $f(x)=\sqrt{3x^2-4x}$
7. $y=\dfrac{1}{\sqrt{x^2+1}}$
8. $y=\sqrt[3]{x^3+8}$
9. $y=\dfrac{1}{\sqrt[3]{x^3+1}}$
10. $f(x)=(1+4x^2)^{-\frac{3}{2}}$

Answers

1. $10(2x+1)^4$
2. $-3(1-x)^2$
3. $2(3x^2-2)(x^3-2x)$
4. $(1-x)^{-2}$
5. $-4(2x+3)^{-3}$
6. $(3x-2)(3x^2-4x)^{-\frac{1}{2}}$
7. $-x(x^2+1)^{-\frac{3}{2}}$
8. $x^2(x^3+8)^{-\frac{2}{3}}$
9. $-x^2(x^3+1)^{-\frac{4}{3}}$
10. $-12x(1+4x^2)^{-\frac{5}{2}}$

STAPLES

THE CHAIN RULE - Exercises

1. Differentiate each of the following:

   a. $y=\left(3+x^3\right)^5$

   b. $y=\left(1-\frac{x}{6}\right)^7$

   c. $y=\frac{1}{4}\left(3x^2-2x+8\right)^{24}$

   d. $y=\frac{1}{(3-4x)^2}$

   e. $y=\frac{1}{4x-x^2}$

   f. $y=\sqrt{3x-2}$

   g. $y=\sqrt[3]{x^2-9}$

   h. $y=\left(ax^2+bx+c\right)^m$

2. Find the equation of the tangent line to the curve $y=\sqrt{7+\sqrt{x}}$ at the point (4, 3).

3. Find $\frac{dy}{dx}$ at the indicated value of x.

   a. $y=2u^3+3u^2$, $u=x+\frac{1}{\sqrt{x}}$, at $x=1$.

   b. $y=\frac{1}{\left(1+u^2\right)^2}$, $u=\sqrt{x}-1$, at $x=4$.

   c. $y=u^5+u^3$, $u=4v+3v^{-1}$, $v=3-x^2$ at $x=2$.

4. Differentiate $f(x)=\frac{2x^2-2x+7}{x-3}$ by using long division to first write the function in the form $f(x)=Ax+B+\frac{C}{x-3}$.

5. Find a formula for $\frac{dy}{dx}$ when $x=\frac{1}{2}$ if:

   a. $y=f\left(x^3\right)$ b. $y=\left[f(x)\right]^3$ c. $y=\left(1+f(x)\right)^2$

6. Let $y=f\left(x^2+3x-5\right)$. Find $\frac{dy}{dx}$ when $x=1$, given that $f'(-1)=2$.

7. Let $p = q(r(t))$ where $r(t) = (t^2 + t + 2)^{-1/3}$. If $q'\left(\frac{1}{2}\right) = 3$, find $\frac{dp}{dt}$ when $t = 2$.

8. Let $y = \sqrt{x + f(x^2 - 1)}$. Find $\frac{dy}{dx}$ when $x = 3$, given that $f(8) = 0$ and $f'(8) = 3$.

Answers

1. **a.** $15x^2(3+x^3)^4$ **b.** $-\frac{7}{6}\left(1-\frac{x}{6}\right)^6$ **c.** $12(3x-1)(3x^2-2x+8)^{23}$

   d. $\frac{8}{(3-4x)^3}$ **e.** $\frac{2(x-2)}{x^2(4-x)^2}$ **f.** $\frac{3}{2\sqrt{3x-2}}$

   g. $\frac{2x}{3\sqrt[3]{x^2-9}^2}$ **h.** $m(2ax+b)(ax^2+bx+c)^{m-1}$

2. $x - 24y + 68 = 0$

3. **a.** 18 **b.** $-\frac{1}{8}$ **c.** -48608

4. $2 - \frac{19}{(x-3)^2}$

5. **a.** $\frac{3}{4}f'\left(\frac{1}{8}\right)$ **b.** $3\left[f\left(\frac{1}{2}\right)\right]^2 f'\left(\frac{1}{2}\right)$ **c.** $2\left(1+f\left(\frac{1}{2}\right)\right)f'\left(\frac{1}{2}\right)$

6. 10

7. $-\frac{5}{16}$

8. $\frac{19\sqrt{3}}{6}$